AP CALCULUS AB

Unit 1 • Worksheet 1-8A • Answer key

Worksheet 1-8A • MVT

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

For $f(x) = x^2$ on $[0, 4]$, which value of c satisfies the MVT?

(A) 1

(B) 2

(C) 3

(D) 4

Solution

Answer: (B). Secant slope $4 = 2c$.

Question 2

NO CALCULATOR

For which function does the MVT apply on $[-1, 1]$?

(A) $|x|$

(B) $\frac{1}{x}$

(C) x^3

(D) $\sqrt[3]{x}$

Solution

Answer: (C). $|x|$ has a corner and $\sqrt[3]{x}$ a vertical tangent at 0; $\frac{1}{x}$ is discontinuous.

Question 3

NO CALCULATOR

f is differentiable, $f(1) = 4$ and $f(5) = -4$. Which must be true?

(A) $f'(c) = -2$ for some c in $(1, 5)$

(B) $f'(3) = -2$

(C) $f(c) = 0$ only once

(D) $f'(c) = 0$ for some c

Solution

Answer: (A). $\frac{-8}{4} = -2$.

Question 4 (Free response) NO CALCULATOR

Find all values of c guaranteed by the MVT. (Use $g'(x) = 3x^2$ and $k'(x) = -\frac{1}{x^2}$.) (a) $g(x) = x^3$ on $[0, 3]$ (b) $k(x) = \frac{1}{x}$ on $[1, 4]$ (c) $g(x) = x^3$ on $[-2, 2]$

Solution
(a) $9 = 3c^2$: $c = \sqrt{3}$. (b) $-\frac{1}{4} = -\frac{1}{c^2}$: $c = 2$. (c) $\frac{8-(-8)}{4} = 4 = 3c^2$: $c = \pm\frac{2}{\sqrt{3}} \approx \pm 1.155$ (both in $(-2, 2)$).

Question 5 (Free response) NO CALCULATOR

A car's position $s(t)$ (miles) is differentiable; $s(0) = 0$, $s(1.5) = 90$, $s(3) = 165$.

(a) Must the car's velocity have been exactly 60 mph at some time in $(0, 1.5)$? Justify.
(b) Must it have been exactly 50 mph at some time in $(1.5, 3)$? Justify.

Solution
(a) $\frac{90}{1.5} = 60$; MVT: yes. (b) $\frac{165-90}{1.5} = 50$; MVT: yes.